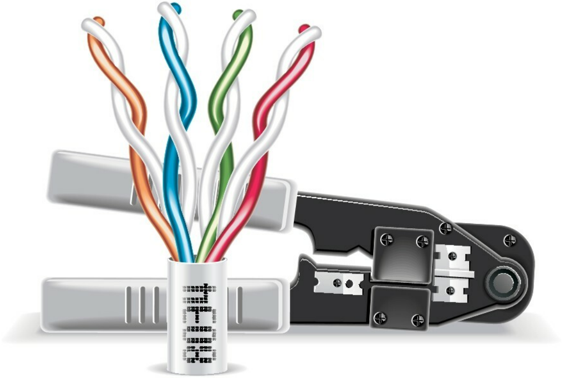

LINUX ON POWER

Anyone familiar with GNU/Linux will not be surprised by the fact that this operating system runs on almost all known processors. However, very few people are aware that mere support just might not be enough. You'll also need to keep an up-to-date repository of code. This is especially true when it comes to serious hardware such as Power.

I guess that almost all desktop systems in the office or home are derived from one prototype, the IBM PC (personal computer), which was built around the concept of an open architecture that IBM introduced 30 years ago. It's true that the company itself didn't quite understand how revolutionary open hardware architecture would be a few years later, in the mid-80s. Whatever the case, the fact remains that the PC became popular because many independent manufacturers could produce a wide range of compatible peripherals. This, in turn, allowed a competitive market to develop, eventually making desktop PCs available to everyone.

The open development model of Linux first embraced the desktop market of x86 computers, and then proceeded to 'capture' other architectures as well (now more than 20), ranging from x86, ARM and MIPS, to Alpha, SPARC, PowerPC, IA64 and S390.

While 'desktop' platforms (x86 and ARM) have many different Linux distributions and other operating systems that can run on them, the 'pure' server architectures like IA64, S390 and Power cannot boast of this—the number of distributions is much, much smaller. This fact is understandable. Quality and the cost of server hardware for industrial applications is immeasurably higher than for the usual PC. Moreover, not every Linux distribution vendor can find the resources to build and provide long-term support for major server hardware architecture—even if they are willing to.

Some time ago, I got my hands on an IBM IntelliStation POWER 285 [1] workstation, considered the closest model to large industrial servers. The main brain of the station is a dual-core POWER 5+ processor. Although announced back in 2005, the system is serious stuff even now. Judge for yourself: 4 GB RAM (max capacity 32 GB), with a disk subsystem that has 2x 146 GB SCSI drives (max disk storage 1.2 TB). It has a self-diagnostic system and memory 'self-healing' technology known as ChipKill, a remote HMC (Hardware Management Console) [2] and a pair of NICs working at 1 Gbit. This system is unbeatable for an entry-level enterprise server and simply forces you to take another respectful look at IBM —Big Blue creates innovative and striking products.

Well, that's enough of an introduction. Let's learn Linux installation and operation on IBM hardware. And later decide how it is possible to effectively exploit Linux on Power.